NEET UG SPECIAL FOCUS

Haloalkanes and Haloarenes

Complete Chapter Tutorial • Reaction Mechanisms • Stereochemistry • Polyhalogen Hazards • Exam-Style Assessment

1. Introduction and Clinical Significance

The replacement of one or more hydrogen atoms in aliphatic or aromatic hydrocarbons by halogen atoms (fluorine, chlorine, bromine, iodine) produces **haloalkanes (alkyl halides)** and **haloarenes (aryl halides)**, respectively. In haloalkanes, the halogen is bonded to an sp^3-hybridised tetrahedral carbon atom, whereas in haloarenes, the halogen is attached directly to an sp^2-hybridised planar aromatic carbon atom.

These organic halides serve as non-polar solvents, versatile synthetic intermediates, and biologically active pharmaceuticals. Several specific compounds highlighted in medical biochemistry and pharmacology frequently appear in NEET UG questions:

- **Chloramphenicol:** A chlorine-containing natural antibiotic synthesized by soil microorganisms; highly effective in the clinical treatment of typhoid fever.
- **Thyroxine:** An iodine-containing hormone secreted by the thyroid gland; deficiency of thyroxine leads to enlargement of the thyroid gland, known as *goiter*.
- **Chloroquine:** A synthetic halogenatedquinoline derivative used worldwide in the prophylaxis and therapy of malaria.
- **Halothane ($CF_3 - CHBrCl$):** A volatile synthetic halogenated hydrocarbon employed widely as a safe, non-explosive general inhalation anaesthetic during surgical procedures.
- **Fluorinated Hydrocarbons:** Several fully fluorinated synthetic compounds are biocompatible with high oxygen-dissolving capacities and are clinically investigated as potential blood substitutes in surgery.

> **Environmental Persistence Note**
>
> Due to the great strength and stability of carbon-halogen bonds, synthetic halogenated compounds resist metabolic degradation by soil bacteria, leading to environmental bioaccumulation and ecological hazards.

2. Classification of Halogen Derivatives

Halogen compounds are classified on two major systematic criteria:

2.1 Classification Based on Number of Halogen Atoms

Depending on whether molecules contain one, two, three, or more halogen atoms, they are classified as **monohalo**, **dihalo**, **trihalo**, or **polyhalo** compounds:

- **Monohaloalkane / Monohaloarene:** e.g., Ethyl bromide (C_2H_5Br), Chlorobenzene (C_6H_5Cl).
- **Dihaloalkane:**
  - *Geminal dihalides (gem-dihalides or alkylidene halides):* Both halogen atoms reside on the *same* carbon atom, e.g., Ethylidene chloride ($CH_3 - CHCl_2$, 1,1-dichloroethane).
  - *Vicinal dihalides (vic-dihalides or alkylene dihalides):* Halogen atoms reside on two *adjacent* carbon atoms, e.g., Ethylene dichloride ($Cl - CH_2 - CH_2 - Cl$, 1,2-dichloroethane).
- **Dihaloarene / Trihaloarene:** e.g., 1,2-dichlorobenzene, 1,3,5-tribromobenzene.

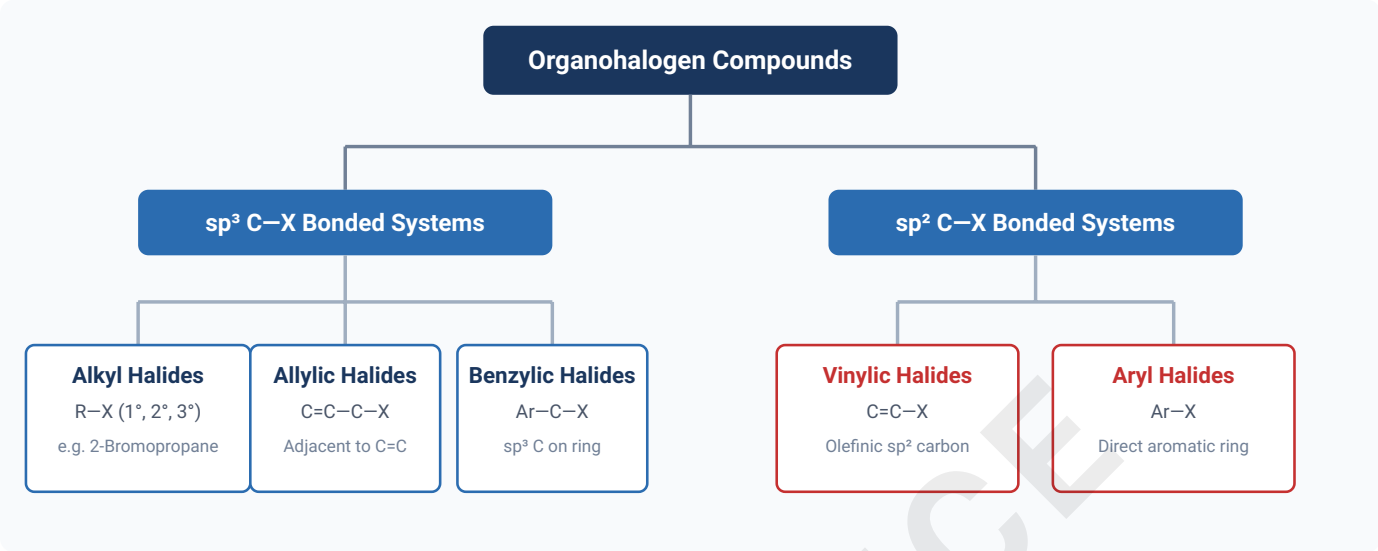
2.2 Classification Based on Hybridisation of the Halogen-Bearing Carbon

Halogen compounds are divided based on whether the carbon bonded to halogen is sp^3- or sp^2-hybridised:

Class	C-X Hybridisation	Structural Description	Representative Example
Alkyl halides (Haloalkanes)	sp^3 $C-X$	Halogen bonded to an alkyl group chain; classified into 1°, 2°, or 3° according to degree of substitution of the α-carbon.	1°: $CH_3CH_2CH_2Br$ 2°: $(CH_3)_2CHBr$ 3°: $(CH_3)_3CBr$
Allylic halides	sp^3 $C-X$	Halogen atom bonded to an sp^3-hybridised carbon adjacent to an aliphatic carbon-carbon double bond ($C=C-C-X$).	Allyl bromide: $CH_2=CH-CH_2Br$ 3-Chlorocyclohex-1-ene
Benzylic halides	sp^3 $C-X$	Halogen atom bonded to an sp^3-hybridised side-chain carbon directly attached to an aromatic ring.	Benzyl chloride: $C_6H_5CH_2Cl$ 2°: $C_6H_5CH(Cl)CH_3$ 3°: $C_6H_5C(Cl)(CH_3)_2$
Vinylic halides	sp^2 $C-X$	Halogen bonded directly to an sp^2-hybridised olefinic carbon of a double bond ($C=C-X$).	Vinyl chloride: $CH_2=CH-Cl$ 1-Chlorocyclohex-1-ene

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Class	C-X Hybridisation	Structural Description	Representative Example
Aryl halides (Haloarenes)	sp^2 C − X	Halogen bonded directly to an sp^2-hybridised aromatic nuclear carbon.	Chlorobenzene: C_6H_5Cl *p*-Chlorotoluene: $CH_3 - C_6H_4 - Cl$



Structural classification scheme based on carbon hybridisation and local bonding environment.

3. IUPAC Nomenclature and Isomerism

In IUPAC nomenclature, alkyl halides are treated as halo-substituted hydrocarbons. Halogen atoms are designated by prefixes (*fluoro-*, *chloro-*, *bromo-*, *iodo-*) arranged alphabetically with appropriate numerical locants indicating position on the longest continuous carbon chain.

Chemical Formula	Common Name	IUPAC Name	Structural Sub-type
$CH_3CH_2CH(Cl)CH_3$	*sec*-Butyl chloride	2-Chlorobutane	Secondary (2°) alkyl halide
$(CH_3)_3CCH_2Br$	*neo*-Pentyl bromide	1-Bromo-2,2-dimethylpropane	Primary (1°) alkyl halide
$(CH_3)_3CBr$	*tert*-Butyl bromide	2-Bromo-2-methylpropane	Tertiary (3°) alkyl halide
$CH_2 = CH - Cl$	Vinyl chloride	Chloroethene	Vinylic halide
$CH_2 = CH - CH_2Br$	Allyl bromide	3-Bromopropene	Allylic halide
$C_6H_5CH_2Cl$	Benzyl chloride	Chlorophenylmethane	Benzylic halide
$o\text{-}Cl - C_6H_4 - CH_3$	*o*-Chlorotoluene	1-Chloro-2-methylbenzene	Aryl halide
CH_2Cl_2	Methylene chloride	Dichloromethane	Polyhaloalkane
$CHCl_3$	Chloroform	Trichloromethane	Trihaloalkane
CCl_4	Carbon tetrachloride	Tetrachloromethane	Tetrahaloalkane

Constitutional Isomers of $C_5H_{11}Br$

Molecular formula $C_5H_{11}Br$ possesses exactly eight constitutional (structural) isomers:

1. $CH_3CH_2CH_2CH_2CH_2Br$: 1-Bromopentane (1°)
2. $CH_3CH_2CH_2CH(Br)CH_3$: 2-Bromopentane (2°)
3. $CH_3CH_2CH(Br)CH_2CH_3$: 3-Bromopentane (2°)
4. $(CH_3)_2CHCH_2CH_2Br$: 1-Bromo-3-methylbutane (1°)
5. $(CH_3)_2CHCH(Br)CH_3$: 2-Bromo-3-methylbutane (2°)
6. $(CH_3)_2C(Br)CH_2CH_3$: 2-Bromo-2-methylbutane (3°)
7. $CH_3CH_2CH(CH_3)CH_2Br$: 1-Bromo-2-methylbutane (1°)
8. $(CH_3)_3CCH_2Br$: 1-Bromo-2,2-dimethylpropane (1°)

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4. Nature of $C-X$ Bond and Physical Properties

Halogens are distinctly more electronegative than carbon. Consequently, the shared pair of electrons in the $C-X$ σ-bond is displaced towards the halogen atom, imparting a partial positive charge (δ^+) to carbon and a partial negative charge (δ^-) to the halogen atom:

$$\overset{\delta^+}{-C}-\overset{\delta^-}{X}$$

Down Group 17 ($F \to Cl \to Br \to I$), atomic size increases progressively while electronegativity decreases. Consequently, the $C-X$ bond length increases, and $C-X$ bond enthalpy decreases.

Halomethane Molecule	Bond Length (pm)	Bond Enthalpy ($kJ \cdot mol^{-1}$)	Dipole Moment (Debye, D)
CH_3-F	139	452	1.847
CH_3-Cl	178	351	**1.860** (Maximum)
CH_3-Br	193	293	1.830
CH_3-I	214	234	1.636

High-Yield NEET Fact: Dipole Moment Anomaly

Dipole moment is the product of charge separation (q) and bond distance (d): $\mu = q \times d$. Although fluorine is more electronegative than chlorine (yielding higher q), the $C-Cl$ bond distance is significantly longer (178 pm vs 139 pm). This greater distance overcomes the slightly smaller charge separation, making the dipole moment of CH_3Cl ($1.860\ D$) strictly greater than that of CH_3F ($1.847\ D$).
Trend: $CH_3Cl > CH_3F > CH_3Br > CH_3I$.

4.1 Boiling Points

- **Physical state at room temperature:** Methyl chloride (CH_3Cl), methyl bromide (CH_3Br), ethyl chloride (CH_3CH_2Cl), and chlorofluoromethanes are gases. Higher members are colourless liquids or solids.
- **Comparison with parent hydrocarbons:** Alkyl halides possess significantly higher boiling points than hydrocarbons of comparable molecular mass because of polar $C-X$ bonds generating strong dipole-dipole interactions in addition to higher van der Waals dispersion forces.
- **Effect of halogen size:** For a given alkyl group, boiling points decrease in the order:
  $$R-I > R-Br > R-Cl > R-F$$
  This occurs because atomic size and polarisable electron clouds increase from F to I, strengthening intermolecular van der Waals forces.
- **Effect of chain branching:** Among isomeric haloalkanes, boiling point *decreases* systematically with increasing branchiness:
  $$\underset{(375\ K)}{CH_3CH_2CH_2CH_2Br} > \underset{(364\ K)}{CH_3CH_2CH(Br)CH_3} > \underset{(346\ K)}{(CH_3)_3CBr}$$
  Branching causes the molecule to adopt a more spherical shape, decreasing total surface area and reducing attractive van der Waals contact area.

4.2 Melting Points and Symmetry of Isomeric Dihalobenzenes

For isomeric dihalobenzenes (e.g., dichlorobenzenes), the boiling points are remarkably close: *ortho* ($453\ K$), *meta* ($446\ K$), and *para* ($448\ K$). However, the **melting point of the *para*-isomer is dramatically higher**:

$$\text{Melting Points: } p\text{-Dichlorobenzene } (323\ K) \gg o\text{-Dichlorobenzene } (256\ K) > m\text{-Dichlorobenzene } (249\ K)$$

Explanation: The symmetrical linear structure of the *para*-isomer allows its molecules to pack tightly and uniformly into the crystal lattice, maximising intermolecular attractions and requiring greater thermal energy to disrupt the crystal solid.

4.3 Density and Solubility

- **Density:** Alkyl fluorides and monochloroalkanes are generally lighter than water. In contrast, alkyl bromides, alkyl iodides, and polyhaloalkanes are denser than water. Density increases with increasing carbon count, number of halogen atoms, and atomic mass of the halogen.
- **Solubility in water:** Haloalkanes are only very slightly soluble (virtually immiscible) in water. Even though they are polar, haloalkanes cannot form hydrogen bonds with water molecules. The energy released when new dipole-dipole attractions form between haloalkane and water molecules is insufficient to overcome both the existing halogen-halogen van der Waals attractions and the strong hydrogen-bonding network between water molecules.
- **Solubility in organic solvents:** Readily soluble in non-polar organic solvents (benzene, ether, alcohol, chloroform) because solute-solvent interactions are similar in strength to the intermolecular forces being broken.

5. Methods of Preparation of Haloalkanes

5.1 From Alcohols

The replacement of the alcoholic hydroxyl ($-OH$) group by a halogen atom is the most common laboratory approach:

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1. Halogen acids (HX):

$$R-OH + HCl \xrightarrow{\text{anhydrous } ZnCl_2} R-Cl + H_2O \quad (\text{Lucas Test})$$
$$R-OH + NaBr + H_2SO_4 \longrightarrow R-Br + NaHSO_4 + H_2O$$

The reactivity of alcohols with a given haloacid follows the order: $3^\circ > 2^\circ > 1^\circ$. Concentrated HCl and anhydrous $ZnCl_2$ (Lucas reagent) reacts with 3° alcohols immediately at room temperature (producing instant cloudiness/turbidity), 2° alcohols react within 5 minutes, while 1° alcohols require heating.

2. Phosphorus halides (PX_3 and PCl_5):

$$3R-OH + PX_3 \longrightarrow 3R-X + H_3PO_3 \quad (X = Cl, Br;\ PBr_3/PI_3 \text{ formed } in\ situ \text{ from red } P + X_2)$$
$$R-OH + PCl_5 \longrightarrow R-Cl + POCl_3 + HCl\uparrow$$

3. Thionyl chloride ($SOCl_2$ - Darzens Procedure):

$$R-OH + SOCl_2 \longrightarrow R-Cl + SO_2\uparrow + HCl\uparrow$$

Why is $SOCl_2$ the Preferred Reagent?

Thionyl chloride ($SOCl_2$) is preferred over phosphorus halides because both co-products, sulfur dioxide (SO_2) and hydrogen chloride (HCl), are escapable gases, leaving the pure alkyl chloride without requiring tedious purification.

Why H_2SO_4 is NEVER used during preparation of Alkyl Iodide from $ROH + KI$

Sulfuric acid is a powerful oxidizing agent. It oxidizes potassium iodide (KI) to hydrogen iodide (HI), and immediately oxidizes the formed HI further into molecular iodine (I_2):

$$2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$$

Thus, HI is destroyed and cannot react with alcohol. Instead, a non-oxidizing acid like **95% orthophosphoric acid (H_3PO_4)** must be used:

$$R-OH + KI + H_3PO_4 \xrightarrow{\Delta} R-I + KH_2PO_4 + H_2O$$

Aryl Halides Cannot Be Prepared from Phenols

Phenols do not react with HX, PCl_5, or $SOCl_2$ to yield aryl halides. In phenols, the unshared electron pair on oxygen is delocalised into the π-system of the aromatic ring, imparting a **partial double-bond character** to the $C-O$ bond, making it extremely strong and resistant to cleavage.

5.2 From Hydrocarbons

1. **Free-radical halogenation of alkanes:** Chlorination or bromination of alkanes in ultraviolet light or heat yields complex mixtures of isomeric mono- and polyhaloalkanes that have close boiling points and are difficult to separate pure.

$$CH_3CH_2CH_2CH_3 \xrightarrow[h\nu]{Cl_2} \underset{\text{(1-chlorobutane)}}{CH_3CH_2CH_2CH_2Cl} + \underset{\text{(2-chlorobutane)}}{CH_3CH_2CH(Cl)CH_3}$$

2. **Electrophilic addition of HX to alkenes:** Follows *Markovnikov's rule* (electrophile H^+ adds to the carbon possessing more hydrogen atoms to yield the more stable carbocation intermediate):

$$CH_3-CH=CH_2 + HI \longrightarrow \underset{\text{Major (2-iodopropane)}}{CH_3-CH(I)-CH_3} + \underset{\text{Minor (1-iodopropane)}}{CH_3-CH_2-CH_2I}$$

   In the presence of organic peroxides, addition of HBr (only HBr, not HCl or HI) occurs via anti-Markovnikov free-radical addition (the Kharasch effect).

3. **Addition of halogens (Test for Unsaturation):** Addition of reddish-brown bromine dissolved in CCl_4 to alkenes forms a colourless vicinal dibromide. The discharge of the reddish-brown colour serves as a standard diagnostic laboratory test for unsaturation.

$$CH_2=CH_2 + Br_2 \xrightarrow{CCl_4} \underset{\text{vic-dibromide (colourless)}}{Br-CH_2-CH_2-Br}$$

5.3 Halogen Exchange Reactions

1. Finkelstein Reaction (Synthesis of Alkyl Iodides):

$$R-X + NaI \xrightarrow{\text{dry acetone}} R-I + NaX\downarrow \quad (X = Cl, Br)$$

Driving Force: Sodium chloride ($NaCl$) and sodium bromide ($NaBr$) are insoluble in dry acetone and precipitate out, shifting the equilibrium in the forward direction according to Le Chatelier's principle.

2. Swarts Reaction (Synthesis of Alkyl Fluorides):

$$CH_3-Br + AgF \xrightarrow{\Delta} CH_3-F + AgBr\downarrow$$

BOSONANCE

Heating alkyl chlorides or bromides with heavy metal fluorides such as AgF, Hg_2F_2, CoF_2, or SbF_3 cleanly produces alkyl fluorides.

6. Methods of Preparation of Haloarenes

6.1 Nuclear Electrophilic Aromatic Substitution of Arenes

Direct chlorination and bromination of benzene or alkylbenzenes occurs smoothly in the presence of Lewis acid catalysts (anhydrous $FeCl_3$, $FeBr_3$, or Fe metal) in the dark:

$$C_6H_5CH_3 + Cl_2 \xrightarrow{\text{Fe, dark}} \underset{\text{o-chlorotoluene}}{\text{o-Cl-}C_6H_4CH_3} + \underset{\text{p-chlorotoluene (Major)}}{\text{p-Cl-}C_6H_4CH_3}$$

The *ortho* and *para* isomers can be easily separated by fractional crystallization because of the substantial difference in their melting points.

- **Iodination:** Aromatic iodination with I_2 is reversible. To drive the reaction forward, it must be performed in the presence of strong oxidising agents like HNO_3 or HIO_4, which oxidize the byproduct HI to I_2.
- **Fluorination:** Fluoroarenes cannot be prepared by this method due to the violent, explosive reactivity of fluorine gas.

6.2 From Diazonium Salts (Sandmeyer Reaction)

When a primary aromatic amine (aniline) dissolved in cold aqueous mineral acid (HCl or HBr) is treated with sodium nitrite ($NaNO_2$) at $273\text{--}278\ K$ ($0\text{--}5^\circ C$), a diazonium salt is formed:

$$C_6H_5NH_2 + NaNO_2 + 2HX \xrightarrow{273\text{--}278\ K} C_6H_5N_2^+X^- + NaX + 2H_2O$$

Mixing the freshly prepared diazonium salt with cuprous chloride (Cu_2Cl_2) or cuprous bromide (Cu_2Br_2) causes replacement of the diazonium group by $-Cl$ or $-Br$, liberating nitrogen gas:

$$C_6H_5N_2^+Cl^- \xrightarrow{Cu_2Cl_2/HCl} C_6H_5Cl + N_2\uparrow \quad \text{(Sandmeyer Reaction)}$$

$$C_6H_5N_2^+Br^- \xrightarrow{Cu_2Br_2/HBr} C_6H_5Br + N_2\uparrow$$

> **Synthesis of Iodobenzene Does Not Require Cuprous Halide!**
>
> Preparation of iodobenzene does not require cuprous iodide (Cu_2I_2). Shaking the diazonium salt directly with aqueous potassium iodide (KI) smoothly produces iodobenzene in high yield:
>
> $$C_6H_5N_2^+Cl^- + KI \longrightarrow C_6H_5I + KCl + N_2\uparrow$$

7. Chemical Reactions of Haloalkanes

Haloalkanes participate in three principal categories of chemical reactions:

1. Nucleophilic substitution reactions
2. Elimination reactions (β-elimination)
3. Reactions with metals (formation of organometallics)

7.1 Nucleophilic Substitution Reactions (S_N)

In a nucleophilic substitution, an electron-rich species (the nucleophile, $Nu:^-$) attacks the partially positive α-carbon, displacing the halogen atom as a halide ion (the leaving group, X^-):

$$Nu:^- + R-X \longrightarrow R-Nu + X^-$$

Reagent	Attacking Nucleophile (Nu^-)	Substitution Product ($R-Nu$)	Class of Organic Product
Aqueous $NaOH$ or KOH	HO^-	ROH	Alcohol
H_2O	H_2O	ROH	Alcohol
Sodium alkoxide ($NaOR'$)	$R'O^-$	ROR'	Ether (Williamson Synthesis)
NaI (in dry acetone)	I^-	$R-I$	Alkyl iodide
Ammonia (NH_3)	NH_3	RNH_2	Primary amine
Primary amine ($R'NH_2$)	$R'NH_2$	$RNHR'$	Secondary amine
KCN (ionic)	$:C\equiv N:^-$ (attacks via C)	$R-CN$	Alkyl cyanide (Nitrile)
$AgCN$ (covalent)	$Ag-C\equiv N:$ (attacks via N)	$R-NC$	Alkyl isocyanide (Isonitrile)
KNO_2 (ionic)	$O=N-O^-$ (attacks via O)	$R-O-N=O$	Alkyl nitrite
$AgNO_2$ (covalent)	$Ag-O-N=O$ (attacks via N)	$R-NO_2$	Nitroalkane
Silver carboxylate ($R'COOAg$)	$R'COO^-$	$R'COOR$	Ester
$LiAlH_4$	H^- (hydride ion)	RH	Hydrocarbon (Alkane)

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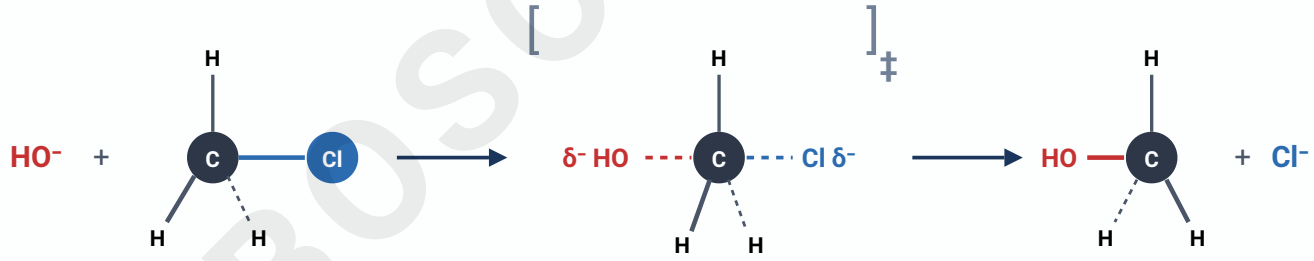
Ambident Nucleophiles in Detail

Ambident nucleophiles are nucleophiles possessing two distinct nucleophilic centres through either of which they can coordinate:

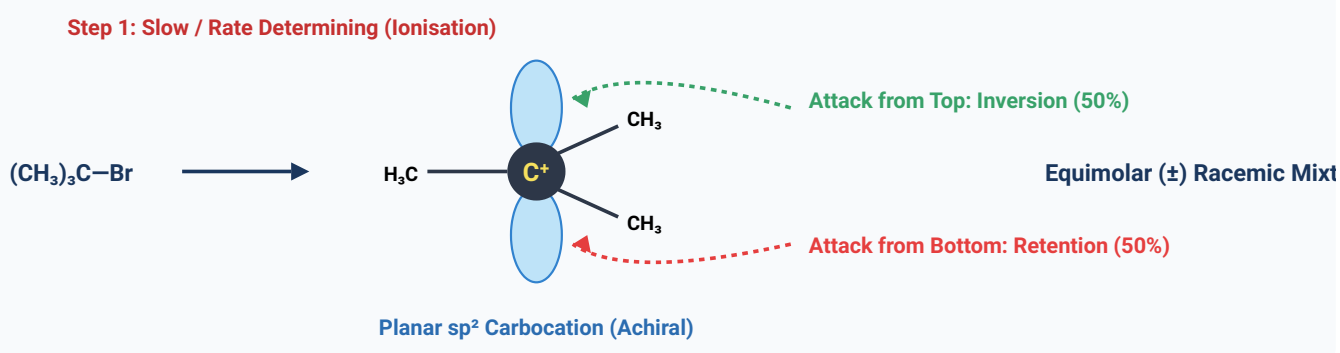
- **Cyanide ion (CN^-):** It is a resonance hybrid: $^-:C\equiv N: \longleftrightarrow :C=\ddot{N}:^-$. With KCN, which is predominantly ionic, cyanide ions are free in solution. Although both C and N can donate electron pairs, attack occurs preferentially through the carbon atom because the resulting $C-C$ bond is thermodynamically much stronger and more stable than a $C-N$ bond. In contrast, $AgCN$ is predominantly covalent; silver is bonded to carbon, leaving the lone pair on *nitrogen* free to initiate nucleophilic attack, yielding **alkyl isocyanide** ($R-NC$) as the primary product.
- **Nitrite ion (NO_2^-):** With ionic KNO_2, the negative charge resides predominantly on oxygen; attack through oxygen generates **alkyl nitrites** ($R-O-N=O$). With covalent $AgNO_2$, the $Ag-O$ bond is covalent, so the lone pair on *nitrogen* attacks the alkyl group, producing **nitroalkanes** ($R-NO_2$).

7.2 Comparative Analysis of S_N2 and S_N1 Reaction Mechanisms

Characteristic	S_N2 (Substitution Nucleophilic Bimolecular)	S_N1 (Substitution Nucleophilic Unimolecular)
Number of Steps	Single concerted step (no intermediate)	Two distinct steps via a carbocation intermediate
Reaction Kinetics	Second order: $\text{Rate}=k[RX][Nu^-]$	First order: $\text{Rate}=k[RX]$ (independent of $[Nu^-]$)
Transition State / Intermediate	Pentacoordinate transition state with partial bonds to incoming Nu and outgoing X	Planar, trigonal sp^2-hybridised carbocation intermediate in Step 1 (rate-determining step)
Stereochemical Outcome	Complete **Inversion of configuration** (Walden inversion)	**Racemisation** (with partial net inversion due to ion-pair shielding)
Substrate Reactivity Order	$CH_3X > 1^\circ > 2^\circ \gg 3^\circ$ (dominated by steric hindrance)	$3^\circ > 2^\circ \gg 1^\circ > CH_3X$ (dominated by carbocation stability)
Allylic & Benzylic Halides	High reactivity (unhindered back-side approach)	Extremely high reactivity due to resonance stabilisation of the carbocation
Solvent Preference	Polar aprotic solvents (DMF, DMSO, acetone)	Polar protic solvents (H_2O, ROH, CH_3COOH)
Nature of Nucleophile	Favoured by strong, charged nucleophiles in high concentration	Favoured by weak, neutral nucleophiles (H_2O, ROH)
Leaving Group Ability	$R-I > R-Br > R-Cl \gg R-F$	$R-I > R-Br > R-Cl \gg R-F$



Bimolecular nucleophilic substitution (S_N2): Concerted back-side attack, pentacoordinate planar transition state, and complete Walden inversion of configuration.



Unimolecular substitution (S_N1): Formation of an achiral planar trigonal carbocation intermediate yielding racemisation via attack from either face.

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7.3 Stereochemical Principles of Nucleophilic Substitution

- **Plane Polarised Light and Optical Activity:** Ordinary light vibrates in all planes perpendicular to the direction of propagation. Passing it through a Nicol prism produces plane-polarised light vibrating in only one plane. Compounds that rotate this plane are **optically active**. Clockwise rotation is termed **dextrorotatory** (designated by $+$ or d), while anticlockwise rotation is termed **laevorotatory** (designated by $-$ or l).
- **Chirality and Stereocentres:** A tetrahedral carbon bonded to four distinct atoms or groups is called an **asymmetric carbon** or **stereocentre**. A molecule that is non-superimposable on its mirror image lacks reflection symmetry and is termed **chiral**. Chirality is the essential condition for optical activity.
- **Enantiomers:** Stereoisomers related to each other as non-superimposable mirror images. They possess identical physical properties (boiling point, melting point, density, refractive index) but rotate plane-polarised light by equal magnitudes in opposite directions.
- **Racemic Modification (Racemate):** An equimolar (50:50) mixture of two enantiomers has zero net optical rotation due to *external compensation* (the rotation caused by molecules of one isomer is exactly cancelled by the equal and opposite rotation of the other isomer). Represented by the prefix $(\pm)$ or dl, e.g., $(\pm)$-butan-2-ol. The process of converting an enantiomer into a racemic mixture is termed **racemisation**.
- **Retention vs. Inversion:** If no bond to the stereocentre is broken during a reaction, the relative spatial arrangement is preserved (**retention of configuration**). In $\mathrm{S_N2}$ reactions, attack from the side strictly opposite to the leaving group results in 100% **inversion of configuration** (e.g., $(-)$-2-bromooctane reacts with NaOH to produce $(+)$-octan-2-ol). In $\mathrm{S_N1}$ reactions, nucleophilic attack on the flat carbocation from both faces yields **racemisation**.

7.4 Elimination Reactions (β-Elimination and Saytzeff Rule)

When a haloalkane containing at least one β-hydrogen is heated with a strong base such as alcoholic potassium hydroxide ($\mathrm{alc.\ KOH}$), a proton is removed from the β-carbon and the halogen departs from the α-carbon, generating an alkene:

$$\mathrm{R-CH_2-CH_2-X + KOH(alc.) \xrightarrow{\Delta} R-CH=CH_2 + KX + H_2O}$$

Saytzeff (Zaitsev) Rule: When dehydrohalogenation can yield more than one isomeric alkene, the preferred major product is the *more highly substituted alkene* (i.e., the alkene carrying the greater number of alkyl substituents attached to the doubly bonded carbons):

$$\underset{\text{2-Bromopentane}}{\mathrm{CH_3-CH_2-CH_2-CH(Br)-CH_3}} \xrightarrow{\text{alc. KOH, }\Delta} \underset{\text{Pent-2-ene (81\% - Major)}}{\mathrm{CH_3-CH_2-CH=CH-CH_3}} + \underset{\text{Pent-1-ene (19\% - Minor)}}{\mathrm{CH_3-CH_2-CH_2-CH=CH_2}}$$

> **Competition: Elimination vs Substitution**
>
> Alkyl halides with β-hydrogens encounter competition between substitution and elimination:
>
> - **Aqueous KOH:** Water has a high dielectric constant and hydrates hydroxide ions ($\mathrm{OH^-}$), reducing their basicity while retaining their nucleophilicity. Hence, **substitution (alcohol formation)** predominates.
> - **Alcoholic KOH:** In alcohol, potassium ethoxide ($\mathrm{C_2H_5O^-K^+}$) forms in equilibrium. Ethoxide is a much stronger base than hydroxide and abstracts a β-proton preferentially, making **elimination (alkene formation)** the major pathway.
> - **Steric Factors:** Bulky bases (e.g., potassium *tert*-butoxide) and sterically hindered 3° halides strongly favor elimination over substitution.

7.5 Reactions with Metals

1. Formation of Grignard Reagents (Organomagnesium Halides):

$$\mathrm{R-X + Mg \xrightarrow{\text{dry ether}} R-Mg-X}$$

In a Grignard reagent, the $\mathrm{C-Mg}$ bond is covalent but highly polar ($\overset{\delta-}{\mathrm{R}}-\overset{\delta+}{\mathrm{Mg}}\overset{\delta-}{\mathrm{X}}$), while the $\mathrm{Mg-X}$ bond is ionic.

Absolute Requirement for Anhydrous Conditions: Grignard reagents are exceptionally reactive bases. They react rapidly with any proton source (water, alcohols, amines, carboxylic acids) to produce alkanes:

$$\mathrm{R-Mg-X + H_2O \longrightarrow R-H + Mg(OH)X}$$

Even traces of atmospheric moisture destroy the reagent. Consequently, Grignard reactions must strictly be conducted under dry ether conditions.

2. Wurtz Reaction:

$$\mathrm{2R-X + 2Na \xrightarrow{\text{dry ether}} R-R + 2NaX}$$

Couples two alkyl radicals to produce symmetrical alkanes having twice the carbon count of the starting alkyl halide.

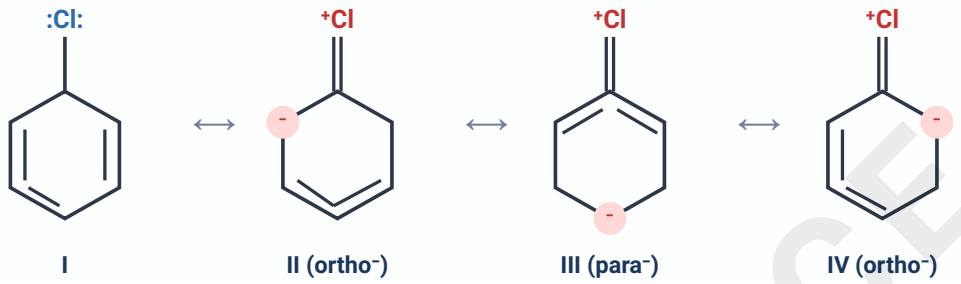
8. Chemical Reactions of Haloarenes

8.1 Low Reactivity of Haloarenes Towards Nucleophilic Substitution

Aryl halides are exceptionally unreactive toward nucleophilic substitution compared to haloalkanes. Four distinct electronic and steric factors account for this stability:

BOSONANCE

1. **Resonance Effect:** The unshared electron pairs on the halogen atom enter into conjugation with the π-electrons of the benzene ring. Resonating canonical structures confer a *partial double-bond character* upon the carbon-halogen ($\mathrm{C-X}$) bond. Cleaving this bond requires significantly more energy than cleaving a pure single bond in haloalkanes.
2. **Difference in Carbon Hybridisation:** In haloalkanes, the halogen is attached to an sp^3-hybridised carbon (25% s-character), whereas in haloarenes, it is bonded to an sp^2-hybridised ring carbon (33.3% s-character). Greater s-character makes the carbon atom more electronegative, drawing the bonding electrons more tightly toward the nucleus. Thus, the $\mathrm{C-Cl}$ bond length in chlorobenzene is shorter ($169\ \mathrm{pm}$) and stronger than that in chloroethane ($177\ \mathrm{pm}$).
3. **Instability of Phenyl Cation:** If self-ionisation were to occur, the resulting phenyl cation ($\mathrm{C_6H_5^+}$) would have its positive charge residing in an sp^2 orbital perpendicular to the π-cloud, precluding any resonance stabilisation. Hence, the $\mathrm{S_N1}$ mechanism is completely ruled out.
4. **Electrostatic Repulsion:** The electron-rich incoming nucleophile experiences repulsive forces as it approaches the π-electron clouds of the aromatic ring.



Resonance canonical forms of chlorobenzene illustrating partial double-bond character of C—Cl and delocalisation of negative charge specifically into ortho and para positions.

8.2 Nucleophilic Displacement of Halogen: Activating Effect of $-\mathrm{NO_2}$ Groups

Under drastic conditions (Dow Process), chlorobenzene can be hydrolysed to phenol by heating with aqueous NaOH at $623\ \mathrm{K}$ and $300\ \mathrm{atm}$:

$$\mathrm{C_6H_5Cl} \xrightarrow[\text{(ii) } \mathrm{H^+}]{\text{(i) NaOH, 623 K, 300 atm}} \mathrm{C_6H_5OH}$$

However, the introduction of strong electron-withdrawing nitro ($-\mathrm{NO_2}$) groups at ***ortho*- and *para*-positions** dramatically increases the reactivity of haloarenes toward nucleophilic substitution ($\mathrm{S_{NAr}}$):

- ***p*-Nitrochlorobenzene:** Reacts smoothly at $443\ \mathrm{K}$ with dilute aqueous NaOH to give p-nitrophenol.
- **2,4-Dinitrochlorobenzene:** Reacts at $368\ \mathrm{K}$ with aqueous alkali to yield 2,4-dinitrophenol.
- **2,4,6-Trinitrochlorobenzene (Picryl chloride):** Hydrolyses readily upon simply **warming with pure water** to produce 2,4,6-trinitrophenol (picric acid).

> **Why Does $-\mathrm{NO_2}$ Activate Only at *ortho* and *para* Positions?**
>
> The nucleophilic aromatic substitution occurs through the formation of a resonance-stabilised carbanion intermediate (Meisenheimer complex). Examination of the resonance structures reveals that the negative charge is delocalised exclusively to the ***ortho* and *para* carbons** with respect to the halogen. When an electron-withdrawing $-\mathrm{NO_2}$ group is located at an *ortho* or *para* position, the negative charge is directly placed on the carbon attached to the strongly electronegative nitrogen atom, which powerfully disperses the charge into its oxygen atoms:
>
> $$>\mathrm{C^--N^+(=O)-O^-} \longleftrightarrow >\mathrm{C=N^+(-O^-)-O^-}$$
>
> In contrast, when the $-\mathrm{NO_2}$ group is at the ***meta*-position**, none of the resonating canonical structures ever bears the carbanionic negative charge on the carbon carrying the nitro group. Consequently, a *meta*-nitro group cannot directly stabilise the negative charge through resonance, exerting virtually no activating effect on nucleophilic substitution.

8.3 Electrophilic Aromatic Substitution Reactions of Haloarenes

Halogen atoms attached to an aromatic ring present an interesting duality:

- **Orientation:** Halogens are ***ortho, para*-directing** because their $+R$ (resonance) electron-donating effect increases electron density specifically at the *ortho* and *para* positions relative to the *meta* position.
- **Reactivity:** Halogens are **net deactivating** because their powerful $-I$ (inductive) electron-withdrawing effect withdraws overall electron density from the ring. Consequently, electrophilic aromatic substitutions on haloarenes proceed more slowly and require more forcing conditions than those on benzene.
- **Steric Predominance:** Because of steric hindrance at the *ortho* position, the ***para*-isomer is overwhelmingly the major product**.

BOSONANCE

Reaction	Reagents & Conditions	Major Product	Minor Product
Halogenation	$Cl_2/\text{anhydrous } FeCl_3$	1,4-Dichlorobenzene (*p*-isomer)	1,2-Dichlorobenzene (*o*-isomer)
Nitration	Conc. HNO_3 + Conc. H_2SO_4	1-Chloro-4-nitrobenzene (*p*-isomer)	1-Chloro-2-nitrobenzene (*o*-isomer)
Sulphonation	Conc. H_2SO_4, Δ	4-Chlorobenzenesulfonic acid (*p*-isomer)	2-Chlorobenzenesulfonic acid (*o*-isomer)
Friedel-Crafts Alkylation	$CH_3Cl/\text{anhydrous } AlCl_3$	1-Chloro-4-methylbenzene (*p*-isomer)	1-Chloro-2-methylbenzene (*o*-isomer)
Friedel-Crafts Acylation	$CH_3COCl/\text{anhydrous } AlCl_3$	4-Chloroacetophenone (*p*-isomer)	2-Chloroacetophenone (*o*-isomer)

8.4 Reaction of Haloarenes with Metals

1. Wurtz-Fittig Reaction:

$$Ar - X + 2Na + R - X \xrightarrow{\text{dry ether}} Ar - R + 2NaX$$

A mixture of an alkyl halide and an aryl halide treated with sodium in dry ether produces an alkylarene (e.g., bromobenzene + bromomethane + Na gives toluene).

2. Fittig Reaction:

$$2Ar - X + 2Na \xrightarrow{\text{dry ether}} Ar - Ar + 2NaX$$

Two molecules of aryl halide couple in the presence of sodium metal in dry ether to yield a biaryl (e.g., chlorobenzene yields diphenyl / biphenyl).

9. Polyhalogen Compounds and Environmental Impact

Carbon compounds containing more than one halogen atom are designated as **polyhalogen compounds**. Many possess significant industrial, pharmaceutical, and agricultural utility, but several pose grave toxicological and environmental hazards.

9.1 Dichloromethane (Methylene Chloride, CH_2Cl_2)

- **Applications:** Widely employed as an industrial solvent for paint removal, as an aerosol propellant, as a process solvent in drug manufacture, and as a metal degreaser.
- **Toxicology:** Highly hazardous to the human central nervous system (CNS). Exposure to lower levels in air induces slight hearing and visual impairment; higher concentrations cause dizziness, nausea, paresthesia, and numbness in extremities. Direct ocular contact burns the cornea.

9.2 Trichloromethane (Chloroform, $CHCl_3$)

- **Applications:** Solvent for fats, alkaloids, and iodine. Extensively consumed in the industrial synthesis of Freon refrigerant $R - 22$. Historically used as a general inhalation anaesthetic.
- **Physiological Effects:** CNS depressant. Short exposure to $900\ \text{ppm}$ causes dizziness, fatigue, and headache. Chronic exposure induces serious hepatic and renal toxicity because chloroform is metabolised in the liver to **phosgene**.
- **Oxidation and Storage:** Chloroform oxidises slowly in the presence of air and light to produce **carbonyl chloride (phosgene, $COCl_2$)**, an intensely poisonous gas:

$$2CHCl_3 + O_2 \xrightarrow{\text{light}} 2COCl_2 + 2HCl$$

Safety storage requirement: Chloroform must strictly be stored in completely filled, airtight, dark amber bottles to completely exclude light and air.

9.3 Triiodomethane (Iodoform, CHI_3)

- **Properties & Use:** Bright yellow crystalline solid with a persistent, pungent medicinal odour. Formerly utilized as an antiseptic dressing for wounds.
- **Key Fact:** The antiseptic action is due entirely to the **liberation of free elemental iodine** upon contact with tissues, not due to the intact iodoform molecule itself. Due to its obnoxious odour, it has been replaced by modern iodophor formulations.

9.4 Tetrachloromethane (Carbon Tetrachloride, CCl_4)

- **Applications:** Feedstock for chlorofluorocarbons (CFCs), industrial solvent for degreasing, and historically used as a commercial fire extinguisher under the trade name *Pyrene*.
- **Health Hazards:** Exposure causes liver cancer, dizziness, nausea, irreversible nerve cell damage, coma, and fatal cardiac arrhythmias.
- **Environmental Impact:** Upon atmospheric release, volatile CCl_4 diffuses into the stratosphere, where photolytic breakdown generates reactive chlorine radicals that catalyze the catalytic destruction of the **stratospheric ozone layer**, leading to elevated terrestrial UV exposure.

9.5 Freons (Chlorofluorocarbons, CFCs)

- **Definition & Properties:** Chlorofluorocarbon derivatives of methane and ethane. Highly stable, unreactive, non-corrosive, non-toxic, easily liquefiable gases.
- **Freon-12 (CCl_2F_2):** Manufactured commercially from carbon tetrachloride via the Swarts reaction:

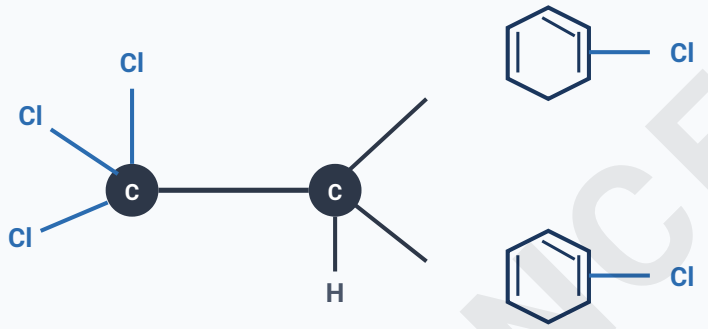
BOSONANCE

$$CCl_4 + 2SbF_3 \xrightarrow{SbCl_5} CCl_2F_2 + 2SbCl_3$$

- **Atmospheric Depletion:** Released freons diffuse unchanged through the troposphere into the stratosphere, where intense UV radiation photolyses $C-Cl$ bonds to generate chlorine free radicals ($Cl^{\bullet}$), which initiate destructive catalytic ozone breakdown cycles.

9.6 p,p'-Dichlorodiphenyltrichloroethane (DDT)

- **History:** First synthesized in 1873. Its powerful insecticidal efficacy was discovered in 1939 by Swiss chemist **Paul Müller** (who was awarded the Nobel Prize in Physiology or Medicine in 1948). Extensively used during and after World War II to combat malaria mosquitoes and typhus-carrying lice.
- **Structure:** 1,1,1-trichloro-2,2-bis(4-chlorophenyl)ethane.
- **Ecological Toxicity & Ban:** Highly stable chemically and exceptionally fat-soluble. Animals metabolise DDT very slowly; it undergoes **biomagnification** through the trophic food web, accumulating in fatty adipose tissues. In birds, DDT disrupts calcium metabolism, causing severe eggshell thinning and population collapse. Consequently, DDT was formally banned in the United States in 1973.



Chemical structure of p,p'-DDT: 2,2-bis(4-chlorophenyl)-1,1,1-trichloroethane.

10. In-Depth Worked Examples

Worked Example 1: Rationalizing S_N2 vs S_N1 Reactivity Profiles

Problem: Arrange the four isomeric bromobutanes in order of increasing reactivity toward:
(a) S_N2 displacement
(b) S_N1 solvolysis

Solution:
The four constitutional isomers of C_4H_9Br are:

- $CH_3CH_2CH_2CH_2Br$ (1-Bromobutane, primary, unbranched at β)
- $(CH_3)_2CHCH_2Br$ (1-Bromo-2-methylpropane, primary, β-branched)
- $CH_3CH_2CH(Br)CH_3$ (2-Bromobutane, secondary)
- $(CH_3)_3CBr$ (2-Bromo-2-methylpropane, tertiary)

(a) S_N2 Reactivity Order:
Controlled by steric hindrance around the electrophilic carbon. Increasing alkyl bulk dramatically blocks back-side nucleophilic approach:

$$(CH_3)_3CBr < CH_3CH_2CH(Br)CH_3 < (CH_3)_2CHCH_2Br < CH_3CH_2CH_2CH_2Br$$

Note: 1-Bromobutane reacts faster than isobutyl bromide because isobutyl bromide possesses a bulky methyl substituent on the adjacent β-carbon.

(b) S_N1 Reactivity Order:
Controlled by carbocation intermediate stability: $3^\circ > 2^\circ > 1^\circ$:

$$CH_3CH_2CH_2CH_2Br < (CH_3)_2CHCH_2Br < CH_3CH_2CH(Br)CH_3 < (CH_3)_3CBr$$

The tertiary carbocation $(CH_3)_3C^+$ has 9 hyperconjugative α-hydrogens and maximum $+I$ dispersal, forming most rapidly.

BOSONANCE

Worked Example 2: Stereochemical Trajectory of Substitution

Problem: An optically pure sample of $(-)$-2-bromooctane is reacted with aqueous sodium hydroxide.
(a) Predict the stereochemical outcome and optical activity of the resulting octan-2-ol.
(b) Explain how the mechanism dictates this outcome.

Solution:
(a) Outcome: The product formed is pure $(+)$**-octan-2-ol**. The reaction proceeds with **complete inversion of configuration**, and the product is optically active.

(b) Mechanism: Because 2-bromooctane is an uncrowded secondary halide and hydroxide ion (OH^-) is a strong nucleophile, the reaction proceeds predominantly through the S_N2 **mechanism**. The hydroxide ion attacks the chiral carbon from the side exactly opposite (at 180°) to the departing bromide leaving group. During the transition state, the three remaining groups ($-H$, $-CH_3$, $-C_6H_{13}$) pass through a planar arrangement, inverting to the opposite side like an umbrella caught in a gale. Consequently, $(-)$-2-bromooctane yields exclusively $(+)$-octan-2-ol.

Worked Example 3: Multi-Step Organic Transformation Scheme

Problem: Identify the missing reagents and structural intermediates A, B, C, D in the following sequential synthetic series:

$$\text{Cyclohexyl bromide} \xrightarrow{\text{Mg, dry ether}} \mathbf{A} \xrightarrow{H_2O} \mathbf{B}$$
$$\mathbf{C} \xrightarrow{\text{Mg, dry ether}} \mathbf{D} \xrightarrow{D_2O} CH_3 - CH(D) - CH_3$$

Solution:
1. Reaction of cyclohexyl bromide with magnesium in dry ether forms the Grignard reagent:
$$\mathbf{A} = \text{Cyclohexylmagnesium bromide } (C_6H_{11}MgBr)$$
2. Hydrolysis of cyclohexylmagnesium bromide with water donates a proton to the nucleophilic cyclohexyl ring:
$$\mathbf{B} = \text{Cyclohexane } (C_6H_{12})$$
3. The deuterated product is 2-deuteropropane ($CH_3 - CH(D) - CH_3$). Since deuterium (D) comes from D_2O, the organometallic intermediate $\mathbf{D}$ must have its $-MgX$ bond at carbon-2:
$$\mathbf{D} = \text{Isopropylmagnesium halide } ((CH_3)_2CH - MgX)$$
4. Therefore, the precursor alkyl halide $\mathbf{C}$ must be:
$$\mathbf{C} = \text{2-Bromopropane or 2-Halopropane } ((CH_3)_2CH - Br)$$

11. NEET UG Exam-Style Practice Questions

The following 20 original practice questions rigorously mirror the latest NEET UG examination format, featuring single-correct multiple-choice questions, Assertion-Reason pairs, Statement-I/II evaluations, and Matrix-Matching problems. Complete worked step-by-step solutions follow each question.

Question 1 (Single Correct MCQ)

Which of the following compounds has the highest dipole moment?

(A) CH_3F

(B) CH_3Cl

(C) CH_3Br

(D) CH_3I

Correct Answer: (B) CH_3Cl
Worked Solution: Dipole moment is given by $\mu = q \times d$. While the electronegativity of fluorine is greater than that of chlorine, the $C - Cl$ bond distance ($178\,pm$) is substantially larger than the $C - F$ bond distance ($139\,pm$). The product of charge and distance yields $\mu = 1.860\,D$ for CH_3Cl and $1.847\,D$ for CH_3F. The exact experimental dipole moments are: CH_3Cl ($1.860\,D$) > CH_3F ($1.847\,D$) > CH_3Br ($1.830\,D$) > CH_3I ($1.636\,D$).

BOSONANCE

Question 2 (Single Correct MCQ)

In the preparation of alkyl iodides from alcohols using KI, which of the following acids is employed instead of concentrated sulfuric acid?

(A) Dilute HCl

(B) Concentrated $\mathrm{HNO_3}$

(C) 95% $\mathrm{H_3PO_4}$

(D) Glacial acetic acid

Correct Answer: (C) 95% $\mathrm{H_3PO_4}$
Worked Solution: Sulfuric acid is a strong oxidising agent; it oxidises KI to HI and immediately oxidises the resulting HI to elemental iodine ($\mathrm{I_2}$). Therefore, non-oxidising phosphoric acid (95% $\mathrm{H_3PO_4}$) must be used to generate HI safely in solution for the nucleophilic conversion of alcohol to alkyl iodide.

Question 3 (Assertion-Reason)

Assertion (A): *para*-Dichlorobenzene has a significantly higher melting point than its *ortho*- and *meta*-isomers.
Reason (R): The *para*-isomer is symmetrical and fits closely into the crystal lattice compared to the *ortho*- and *meta*-isomers.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
Worked Solution: The melting point of *p*-dichlorobenzene is $323\,\mathrm{K}$, whereas those of the *ortho* and *meta* isomers are $256\,\mathrm{K}$ and $249\,\mathrm{K}$, respectively. Due to its linear centrosymmetric structure, the *para*-isomer packs much more efficiently and closely into the crystal lattice, maximising lattice energy and requiring a substantially higher temperature to melt.

Question 4 (Single Correct MCQ)

The reaction of ethyl chloride with potassium cyanide (KCN) and silver cyanide (AgCN) gives predominantly which major products, respectively?

(A) Ethyl isocyanide and ethyl cyanide

(B) Ethyl cyanide and ethyl isocyanide

(C) Ethyl cyanide and ethyl cyanide

(D) Ethyl isocyanide and ethyl isocyanide

Correct Answer: (B) Ethyl cyanide and ethyl isocyanide
Worked Solution: KCN is an ionic compound and yields free cyanide ions; attack occurs preferentially through the carbon atom because $\mathrm{C-C}$ bonds are stronger than $\mathrm{C-N}$ bonds, forming propanenitrile (ethyl cyanide, $\mathrm{CH_3CH_2CN}$). Conversely, AgCN is primarily covalent, with the carbon atom bonded to silver; hence, the unshared pair on the nitrogen atom attacks, producing ethyl isocyanide ($\mathrm{CH_3CH_2NC}$).

BOSONANCE

Question 5 (Statement Evaluation)

Statement I: Hydrolysis of $(-)$-2-bromooctane by aqueous sodium hydroxide proceeds with complete inversion of optical configuration.
Statement II: $\mathrm{S_N2}$ reactions proceed via a planar carbocation intermediate.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C) Statement I is correct but Statement II is incorrect.
Worked Solution: Statement I is correct: alkaline hydrolysis of $(-)$-2-bromooctane occurs via the $\mathrm{S_N2}$ mechanism yielding $(+)$-octan-2-ol with Walden inversion. Statement II is incorrect: $\mathrm{S_N2}$ reactions proceed through a single concerted step via a pentacoordinate transition state without forming any carbocation intermediate. Planar carbocations are formed in $\mathrm{S_N1}$ reactions.

Question 6 (Single Correct MCQ)

Which of the following alkyl halides undergoes $\mathrm{S_N1}$ hydrolysis at the fastest rate?

(A) $\mathrm{C_6H_5-CH_2-Cl}$

(B) $\mathrm{C_6H_5-CH(Cl)-CH_3}$

(C) $\mathrm{C_6H_5-CH(Cl)-C_6H_5}$

(D) $\mathrm{C_6H_5-C(Cl)(CH_3)-C_6H_5}$

Correct Answer: (D) $\mathrm{C_6H_5-C(Cl)(CH_3)-C_6H_5}$
Worked Solution: The rate of an $\mathrm{S_N1}$ reaction is dictated by the stability of the intermediate carbocation formed in the rate-determining step. Compound (D) generates a tertiary benzylic carbocation:

$$[\mathrm{C_6H_5-\overset{+}{C}(CH_3)-C_6H_5}]$$

This carbocation is resonance-stabilised by two phenyl rings as well as hyperconjugative and inductive stabilisation by the methyl group, making it exceptionally stable.

Question 7 (Single Correct MCQ)

When 2-bromopentane is heated with alcoholic potassium hydroxide, the major alkene produced is:

(A) Pent-1-ene

(B) *cis*-Pent-2-ene and *trans*-Pent-2-ene (predominantly 2-pentene)

(C) 2-Methylbut-2-ene

(D) Pent-2-yne

Correct Answer: (B) *cis*-Pent-2-ene and *trans*-Pent-2-ene (predominantly 2-pentene)
Worked Solution: Dehydrohalogenation of 2-bromopentane follows Saytzeff's rule. Elimination of β-hydrogen from $\mathrm{C3}$ produces pent-2-ene (an internal, disubstituted alkene; 81% yield), whereas elimination from $\mathrm{C1}$ yields pent-1-ene (a monosubstituted terminal alkene; 19% yield). Thus, pent-2-ene is the major alkene.

BOSONANCE

Question 8 (Assertion-Reason)

Assertion (A): Chlorobenzene is significantly less reactive than chloroethane towards nucleophilic substitution reactions.
Reason (R): In chlorobenzene, the $\mathrm{C-Cl}$ bond acquires partial double-bond character due to resonance.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
Worked Solution: The lone pairs on chlorine conjugate with the π-electrons of the benzene ring, imparting partial double-bond character to the $\mathrm{C-Cl}$ bond. The bond length is shorter ($169\ \mathrm{pm}$ vs $177\ \mathrm{pm}$) and bond cleavage requires significantly higher energy. This, together with the sp^2 hybridisation of the carbon and phenyl cation instability, renders chlorobenzene inert to normal nucleophilic substitutions.

Question 9 (Matching Matrix)

Match the reactions listed in **List-I** with the corresponding name/reagents in **List-II**:

List-I	List-II
P. $\mathrm{R-Br + NaI \xrightarrow{acetone} R-I + NaBr}$	1. Swarts reaction
Q. $\mathrm{CH_3Br + AgF \longrightarrow CH_3F + AgBr}$	2. Fittig reaction
R. $\mathrm{C_6H_5Cl + CH_3Cl + 2Na \xrightarrow{ether} C_6H_5CH_3}$	3. Finkelstein reaction
S. $\mathrm{2C_6H_5Cl + 2Na \xrightarrow{ether} C_6H_5 - C_6H_5}$	4. Wurtz-Fittig reaction

(A) P-3, Q-1, R-4, S-2

(B) P-1, Q-3, R-4, S-2

(C) P-3, Q-1, R-2, S-4

(D) P-4, Q-2, R-1, S-3

Correct Answer: (A) P-3, Q-1, R-4, S-2
Worked Solution:

- $\mathrm{R-X + NaI \rightarrow R-I}$ is the **Finkelstein reaction** (P-3).
- $\mathrm{R-X + AgF \rightarrow R-F}$ is the **Swarts reaction** (Q-1).
- Aryl halide + alkyl halide + Na is the **Wurtz-Fittig reaction** (R-4).
- Two aryl halides + Na is the **Fittig reaction** (S-2).

BOSONANCE

Question 10 (Single Correct MCQ)

Which of the following compounds will undergo nucleophilic substitution ($\mathrm{S_{NAr}}$) by aqueous sodium hydroxide under the mildest conditions (lowest temperature)?

(A) 4-Chloronitrobenzene

(B) Chlorobenzene

(C) 2,4,6-Trinitrochlorobenzene

(D) 2,4-Dinitrochlorobenzene

Correct Answer: (C) 2,4,6-Trinitrochlorobenzene
Worked Solution: Nitro groups are strongly electron-withdrawing ($-M$ and $-I$). In 2,4,6-trinitrochlorobenzene (picryl chloride), three $-\mathrm{NO_2}$ groups occupy all *ortho* and *para* positions, powerfully stabilising the Meisenheimer carbanion intermediate. It hydrolyses upon simply warming with neutral water ($\mathrm{H_2O}$), without needing sodium hydroxide.

Question 11 (Single Correct MCQ)

Chloroform is slowly oxidised by atmospheric oxygen in the presence of sunlight to form an extremely toxic gas known as:

(A) Mustard gas

(B) Phosgene

(C) Tear gas

(D) Lewisite

Correct Answer: (B) Phosgene
Worked Solution: The photo-oxidation of chloroform is represented by:

$$2\mathrm{CHCl_3} + \mathrm{O_2} \xrightarrow{h\nu} 2\mathrm{COCl_2} + 2\mathrm{HCl}$$

Carbonyl chloride ($\mathrm{COCl_2}$) is phosgene, an insidious respiratory poison.

Question 12 (Statement Evaluation)

Statement I: Allyl chloride and benzyl chloride exhibit high reactivity toward nucleophilic substitution via the $\mathrm{S_N1}$ pathway.
Statement II: The carbocations derived from allyl and benzyl halides are stabilized by resonance delocalisation.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (A) Both Statement I and Statement II are correct.
Worked Solution: Both statements are completely correct. In $\mathrm{S_N1}$ reactions, allyl chloride forms the allyl carbocation ($\mathrm{CH_2=CH-\overset{+}{C}H_2 \longleftrightarrow \overset{+}{C}H_2-CH=CH_2}$) and benzyl chloride forms the benzyl carbocation ($\mathrm{C_6H_5\overset{+}{C}H_2}$), both of which are strongly stabilised by π-electron resonance delocalisation.

BOSONANCE

Question 13 (Single Correct MCQ)

An alkane of molecular formula $\mathrm{C_5H_{12}}$ on photochemical monochlorination yields exclusively **a single monochloride**. The alkane is:

(A) *n*-Pentane

(B) 2-Methylbutane (Isopentane)

(C) 2,2-Dimethylpropane (Neopentane)

(D) Cyclopentane

Correct Answer: (C) 2,2-Dimethylpropane (Neopentane)
Worked Solution: Neopentane, $\mathrm{(CH_3)_4C}$, has 12 equivalent primary hydrogen atoms attached symmetrically to four identical methyl groups. Replacement of any of these twelve hydrogens yields the exact same product: 1-chloro-2,2-dimethylpropane (neopentyl chloride). (Note: *n*-Pentane gives 3 monochloro isomers; isopentane gives 4 monochloro isomers).

Question 14 (Assertion-Reason)

Assertion (A): Chlorine atom is *ortho*- and *para*-directing in electrophilic aromatic substitution of chlorobenzene.
Reason (R): The resonance effect of chlorine increases the electron density at *ortho* and *para* positions relative to *meta* positions.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
Worked Solution: Although chlorine is overall deactivating due to its strong $-I$ effect, its unshared electron pairs donate into the ring by resonance ($+R$ effect), which specifically places partial negative charges at the *ortho* and *para* positions. Thus, incoming electrophiles attack preferentially at *ortho* and *para* carbons.

Question 15 (Single Correct MCQ)

Which one of the following compounds is **achiral**?

(A) 2-Bromobutane

(B) 2-Chloropentane

(C) Propan-2-ol

(D) 2-Bromopropanoic acid

Correct Answer: (C) Propan-2-ol
Worked Solution: Propan-2-ol, $\mathrm{(CH_3)_2CHOH}$, has two identical methyl groups attached to $\mathrm{C2}$. Since it does not possess four different groups on any carbon, it contains a plane of symmetry and is **achiral** and optically inactive. In (A), (B), and (D), the $\mathrm{C2}$ carbon is bonded to four completely different groups, making them chiral.

BOSONANCE

Question 16 (Single Correct MCQ)

The antiseptic property of iodoform ($\mathrm{CHI_3}$) is due to:

(A) Its strong objectionable odour

(B) The liberation of free elemental iodine

(C) The intact triiodomethane molecule acting on cell membranes

(D) Its oxidation into iodic acid

Correct Answer: (B) The liberation of free elemental iodine
Worked Solution: As explicitly established in the chapter syllabus, iodoform itself is not an active antiseptic; rather, when it comes in contact with living tissue, it slowly decomposes and releases free elemental iodine, which is the actual active germicidal agent.

Question 17 (Single Correct MCQ)

Freon-12 is manufactured on a large industrial scale from carbon tetrachloride by which of the following reactions?

(A) Finkelstein reaction

(B) Wurtz reaction

(C) Swarts reaction

(D) Sandmeyer reaction

Correct Answer: (C) Swarts reaction
Worked Solution: Freon-12 ($\mathrm{CCl_2F_2}$) is manufactured from tetrachloromethane ($\mathrm{CCl_4}$) by reacting it with antimony trifluoride ($\mathrm{SbF_3}$) in the presence of antimony pentachloride catalyst, which is an application of the **Swarts reaction** (fluorine-halogen exchange).

BOSONANCE

Question 18 (Matching Matrix)

Match the halogen-containing compounds in **List-I** with their primary clinical or commercial applications in **List-II**:

List-I	List-II
P. Chloramphenicol	1. Treatment of malaria
Q. Chloroquine	2. Inhalation anaesthetic in surgery
R. Halothane	3. Antibiotic for typhoid fever
S. Thyroxine	4. Thyroid hormone preventing goiter

(A) P-3, Q-1, R-2, S-4

(B) P-1, Q-3, R-2, S-4

(C) P-3, Q-2, R-1, S-4

(D) P-4, Q-1, R-2, S-3

Correct Answer: (A) P-3, Q-1, R-2, S-4
Worked Solution:

- **Chloramphenicol:** Natural chlorine-containing antibiotic for typhoid (P-3).
- **Chloroquine:** Synthetic drug for malaria (Q-1).
- **Halothane:** Volatile halogenated general surgical anaesthetic (R-2).
- **Thyroxine:** Iodine-containing hormone; deficiency causes goiter (S-4).

Question 19 (Single Correct MCQ)

The correct IUPAC name of the compound given below is:

$$\mathrm{CH_3-CH=CH-CH(Br)-CH_3}$$

(A) 2-Bromopent-3-ene

(B) 4-Bromopent-2-ene

(C) 1-Bromo-1-methylbut-2-ene

(D) 4-Bromopent-3-ene

Correct Answer: (B) 4-Bromopent-2-ene
Worked Solution: The longest chain contains 5 carbons and a double bond. In IUPAC rules, double bonds take numbering precedence over halogen substituents. Numbering from the left gives the double bond locant 2 ($\mathrm{C2=C3}$), whereas numbering from the right gives it locant 3. Thus, numbering from the left: $\mathrm{C1(CH_3)-C2H=C3H-C4H(Br)-C5H_3}$. The bromo group is at $\mathrm{C4}$. Hence, the IUPAC name is **4-bromopent-2-ene**.

BOSONANCE

Question 20 (Single Correct MCQ)

Which of the following statements about $\mathrm{p,p'}$-DDT is **INCORRECT**?

(A) Its systematic name is 2,2-bis(4-chlorophenyl)-1,1,1-trichloroethane.

(B) It is highly soluble in water and readily metabolised by fish.

(C) It was banned in the United States in 1973 due to biomagnification and environmental persistence.

(D) Paul Müller was awarded the 1948 Nobel Prize in Medicine for discovering its insecticidal effectiveness.

Correct Answer: (B) It is highly soluble in water and readily metabolised by fish.
Worked Solution: Statement (B) is completely false. DDT is extremely hydrophobic (insoluble in water) and highly lipophilic (fat-soluble). It is not rapidly metabolised by fish or other animals; instead, it accumulates in adipose tissues and biomagnifies across ecological food chains, showing high toxicity toward fish and thinning bird eggshells. Statements (A), (C), and (D) are factually correct.

12. Common NEET Pitfalls & Revision Checklist

Critical Traps to Avoid in NEET UG

- **Dipole Moment Order:** Never assume $\mathrm{CH_3F}$ has the highest dipole moment based solely on fluorine's electronegativity. $\mathrm{CH_3Cl}$ $(1.860\ \mathrm{D})$ is higher than $\mathrm{CH_3F}$ $(1.847\ \mathrm{D})$.
- **Lucas Test:** Lucas reagent is conc. HCl + anhydrous $\mathrm{ZnCl_2}$. Turbidity appears immediately for 3° alcohols, in 5 minutes for 2° alcohols, and only upon heating for 1° alcohols.
- **KCN vs AgCN:** KCN gives cyanides $(\mathrm{R-CN})$; AgCN gives isocyanides $(\mathrm{R-NC})$.
- **KNO_2 vs $AgNO_2$:** $\mathrm{KNO_2}$ gives alkyl nitrites $(\mathrm{R-ONO})$; $\mathrm{AgNO_2}$ gives nitroalkanes $(\mathrm{R-NO_2})$.
- **Aqueous vs Alcoholic KOH:** Aqueous KOH promotes nucleophilic substitution $(\mathrm{R-OH})$; alcoholic KOH promotes β-elimination (alkenes).
- **Grignard Reactions:** Must be carried out under strictly anhydrous ether conditions because any active hydrogen ($\mathrm{H_2O, ROH, RNH_2}$) immediately converts RMgX to $\mathrm{R-H}$.
- **Iodobenzene from Diazonium Salt:** No $\mathrm{Cu_2I_2}$ is required. Simply shaking with aqueous KI produces iodobenzene.
- **Halogen Directing Effect vs Reactivity:** Halogens are *ortho, para*-directing (due to $+R$ resonance effect) but *deactivating* (due to strong $-I$ inductive effect).
- **Melting Point of Dihalobenzenes:** *para*-Dichlorobenzene melts at a much higher temperature than *ortho* or *meta* due to crystal lattice symmetry.

Rapid Revision Checklist

- ☑ Know all IUPAC locant priority rules for haloalkenes and substituted arenes.
- ☑ Master the mechanistic differences between $\mathrm{S_N1}$ (carbocation, two steps, racemisation) and $\mathrm{S_N2}$ (concerted, transition state, inversion).
- ☑ Understand Saytzeff's rule: more substituted alkene = major product.
- ☑ Understand the electronic basis for the inertness of aryl and vinyl halides.
- ☑ Memorize the physiological and environmental impacts of chloroform (phosgene), carbon tetrachloride (ozone depletion), Freon-12, and DDT.

13. High-Yield Reaction Master Table

Substrate	Reagents / Conditions	Primary Product	Reaction Type / Mechanism
$\mathrm{R-OH}$	$\mathrm{SOCl_2}$	$\mathrm{R-Cl + SO_2\uparrow + HCl\uparrow}$	Darzens halogenation
$\mathrm{R-Cl}$ or $\mathrm{R-Br}$	NaI in dry acetone	$\mathrm{R-I + NaCl/NaBr\downarrow}$	Finkelstein reaction ($\mathrm{S_N2}$)
$\mathrm{R-Br}$	AgF	$\mathrm{R-F + AgBr\downarrow}$	Swarts reaction
$\mathrm{R-X}$	$\mathrm{KOH(aq)}$	$\mathrm{R-OH}$	Nucleophilic substitution ($\mathrm{S_N}$)
$\mathrm{R-X}$ (with β-H)	$\mathrm{KOH(alc.)},\ \Delta$	Alkene (Saytzeff product)	β-Elimination (Dehydrohalogenation)
$\mathrm{R-X}$	Mg in dry ether	$\mathrm{R-Mg-X}$	Grignard reagent synthesis
$\mathrm{2\,R-X}$	$\mathrm{2Na}$ in dry ether	$\mathrm{R-R + 2NaX}$	Wurtz coupling

BOSONANCE

Substrate	Reagents / Conditions	Primary Product	Reaction Type / Mechanism
$\mathrm{Ar-X + R-X}$	$\mathrm{2Na}$ in dry ether	$\mathrm{Ar-R + 2NaX}$	Wurtz-Fittig coupling
$\mathrm{2Ar-X}$	$\mathrm{2Na}$ in dry ether	$\mathrm{Ar-Ar + 2NaX}$	Fittig coupling
$\mathrm{ArN_2^+Cl^-}$	$\mathrm{Cu_2Cl_2/HCl}$ or $\mathrm{Cu_2Br_2/HBr}$	$\mathrm{Ar-Cl}$ or $\mathrm{Ar-Br}$	Sandmeyer reaction
$\mathrm{ArN_2^+Cl^-}$	$\mathrm{KI(aq)}$, warm	$\mathrm{Ar-I + N_2\uparrow}$	Direct nucleophilic replacement
$\mathrm{C_6H_5Cl}$	$\mathrm{NaOH,\ 623\,K,\ 300\,atm}$ then $\mathrm{H^+}$	$\mathrm{C_6H_5OH}$ (Phenol)	Dow process ($\mathrm{S_{NAr}}$)
p-Nitrochlorobenzene	$\mathrm{NaOH,\ 443\,K}$ then $\mathrm{H^+}$	p-Nitrophenol	Activated $\mathrm{S_{NAr}}$
$\mathrm{C_6H_5Cl}$	$\mathrm{Cl_2/FeCl_3}$	1,4-Dichlorobenzene (major)	Electrophilic aromatic substitution

BOSONANCE